

# BT505 CURRENT PAPERS SOLVED MIDTERM

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## **Define Fouling?**

Fouling is the accumulation of living or non living materials.

## **Define Aptamers?**

Aptamers are oligonucleotide sequences with a length of about 25–80 bases which have abilities to bind to specific target molecules.

## **What is Biotinylation?**

Biotinylation is the process of covalently attaching biotin to a molecule. It is rapid, specific and is unlikely to disturb the natural function. It is required in carboxylation reactions.

## **What is Glassy Carbon?**

Glass-like carbon, (non-graphitizing), combines glassy and ceramic properties with those of graphite.

- Mechanically stable Impermeable to gases and fluids
- Chemically resistant
- High porosity
- Low background current.

Its low sensitivity to peroxide as well as to other mediators.

## **Briefly describe Biometric Verification?**

Biometric verification is any means by which person a can be uniquely identified by evaluating one or more distinguishing biological traits. These biological identifiers include fingerprints, hand and earlobe geometries, retina patterns, voice prints and written signatures. A biometric identifier known as a retinal scan is used to map the unique patterns of a person's retina. This beam traces a path on the retina. Biometric verification is used to identify people in all sorts of settings, including the following:

- Financial institutions
- Healthcare providers
- Law enforcement agencies
- Other government entities

## **Note on Scaling and Scaling forces?**

The study involves the sizing of the object is called scaling.

- Force due to surface tension scales
- Force due to electrostatics with constant field scales
- Force due to certain magnetic forces scales
- Gravitational forces scale

### **Scaling Factors**

Inertia, speed, resistance, heat capacity.

### **Scaling Laws**

Scaling law is a law that describes the variations of physical quantities with the size of the system.

There are two scaling laws.

- **Scaling in Geometry**

Involves Volume and Surface.

- **Scaling of Phenomenological behavior**

Involves size and material characterization.

### **What is Reporter gene?**

A reporter gene is a gene that researchers attach to a regulatory sequence of another gene of interest in bacteria, cell culture, animals or plants. The function of reporter gene is to help in the detection of target molecule. The reporter genes are marker genes to measure the amount of expression.

### **What is Taste sensation and primary taste sensation?**

#### **Taste Sensation**

Taste is the sensation that occurs in the mouth when a substance reacts chemically with taste receptor cells located on taste buds. Taste determines flavours of foods.

#### **Primary taste sensation**

There are five primary taste sensations, namely: sweet, sour, salty, bitter, and umami.

### **DNA as biorecognition sensor?**

DNA or RNA are the Biorecognition elements in Nucleic acid Biosensors. The detection methods are based on recognition of complementary nucleic acid sequence rapid, simple and economical testing for clinical environmental, food analysis and other fields.

### **What is MEMS?**

Microelectromechanical systems MEMS stands for microelectromechanical systems, which combine tiny mechanical components, such as gears and levers, with microelectronic circuitry.

### **What is CO<sub>2</sub> biosensor?**

Electrochemical carbon dioxide sensors measure electrical current to determine how much CO<sub>2</sub> is present in the air. When CO<sub>2</sub> enters the sensor, it chemically reacts within

a polymer surface, resulting in an electrical charge. The type and amount of electrical charge is then used to determine how much CO<sub>2</sub> is present.

### **Write a note on biosensor in blood glucose?**

Biosensors consist of a biorecognition element and physicochemical transducer providing a measurable signal. In case of glucose biosensors, enzyme glucose oxidase is used as a biorecognition element or bio transducer in some sources converting glucose to gluconic acid. A blood glucose sensor works through a tiny sensor inserted under your skin, usually on your belly or arm. The sensor measures your interstitial glucose level, which is the glucose found in the fluid between the cells. The sensor tests glucose every few minutes. A transmitter wirelessly sends the information to a monitor.

### **Quantum dot better than conventional nanotechnology?**

Quantum dots (QDs) are semiconductor particles a few nanometres in size having optical and electronic properties that differ from larger particles due to quantum mechanics whereas the conventional nanotechnology that take place at dimensions in the nanometre scale are utilised in the design, characterisation, production and application of materials, structures, devices and systems.

### **Application of nano coating?**

Nano-coatings are increasingly used in bio textile applications including bio-filtration materials, medical garments, medical fabrics, textile implants, textile substrates for cell growth and other textile products for medical systems.

### **Nanoporous materials, Nano crystalline materials, Hybrid materials?**

#### **Nanoporous materials**

Nanoporous materials consist of a regular organic or inorganic bulk phase in which a porous structure is present. Nanoporous materials exhibit pore diameters that are most appropriately quantified using units of nanometers. The diameter of pores in nanoporous materials is thus typically 100 nanometers or smaller.

#### **Nanocrystalline material**

A nanocrystalline (NC) material is a polycrystalline material with a crystallite size of only a few nanometers. These materials fill the gap between amorphous materials without any long range order and conventional coarse-grained materials.

#### **Hybrid materials**

Hybrid materials are composites made by synergistic combination of organic and inorganic components at the nanometer or molecular level.

### **What are Assay format?**

Assay formats include direct, indirect and sandwich assays which are often used as a basis for comparison with novel platforms in development and are run manually or semi-automated on 96 well plates where samples are measured in duplicate.

### **Explain features and components of biosensor?**

A biosensor is a device that measures biological or chemical reactions by generating signals proportional to the concentration of an analyte in the reaction.

### **Features of Biosensors**

- It should be highly specific for the analyte.
- The reaction used should be independent of manageable factors like pH, temperature, stirring, etc.
- The response should be linear over a useful range of analyte concentrations.

### **Components of Biosensors**

A biosensor includes two main distinct components like Biological component such as cell, enzyme and a physical component like an amplifier and transducer. The biological component identifies as well as communicates through the analyte for generating a signal that can be sensed through the transducer. The important components of a biosensor are

- A bioreceptor (e.g., enzymes, antibody, microorganism, or cells)
- A transducer of the physicochemical signal
- A signal processor to interpret the information that has been converted.

### **What is DNA field-effect transistor (DNAFET)?**

A DNA field-effect transistor (DNAFET) is a field-effect transistor which uses the field-effect due to the partial charges of DNA molecules to function as a biosensor.

### **Two parts of biosensor?**

A biosensor typically consists of :

1. Bio-receptor (enzyme/antibody/cell/nucleic acid/aptamer)
2. Transducer component (semi-conducting material/nanomaterial)
3. Electronic system which includes a signal amplifier, processor & display.

### **Nano gold is significant of bioremediation field?**

Gold nanoparticles (AuNPs) is one of the most relevant metal nanoparticles with enormous applications in various field of research and industries. The demand for AuNPs is increasing rapidly. Characterization of synthesized AuNPs is important in evaluating the morphological properties of AuNPs since the morphological properties of AuNPs affect their potential use in various applications.

**OR**

Nano gold particles also known as colloidal gold AuNPs ranges from 1-50 nm in size AuNPs do not have any adverse effect on BRE and are highly stable. They can be easily conjugates using thiol chemistry. Large surface area for accommodation of BRE. AuNPs come in different shapes such as, star, rods, sphere and branch depending upon the mode of synthesis. For small (30nm) mono- disperse gold NPs, there is an absorption of light in the range ~450 nm. For larger AUNPs light is absorbed in the range of ~700 nm is reflected yielding a rich red colour.

**What are Calorimetric Sensors?**

Calorimetric sensors will refer to optical sensors involve the measurement of heat that is generated (Calorie: unit of heat Calorimetric: Measurement of heat). These sensors typically utilize thermistors that transform heat generated or loss during a reaction into an electrical signal.

**Write down the applications of biosensors?**

- Common healthcare checking
- Metabolites Measurement
- Screening for sickness
- Insulin treatment
- Clinical psychotherapy & diagnosis of disease
- In Military
- Agricultural, and Veterinary applications
- Drug improvement, offense detection
- Processing & monitoring in Industrial
- Ecological pollution control
- Diagnostic & Clinical
- Industrial & Environmental Applications
- Study & Interaction of Biomolecules
- Development of Drug
- Detection of Crime
- Medical Diagnosis
- Monitoring of Environmental Field
- Quality Control
- Process Control in Industries
- Pharmaceuticals Manufacturer & Organs Replacement

**Types of Biosensors?**

Types on the basis of Bioreceptor element

- Enzyme Biosensors
- Immuno Biosensor
- Cell Biosensor
- DNA Biosensor
- Aptamer Biosensor
- Immuno fragment Biosensor
- Biomimetic Biosensor

**Differentiate between mono and poly-clonal antibodies?**

**Monoclonal antibody** is produced by the progeny of a single B cell against single epitope.

**Polyclonal antibody** is generated against multiple epitopes.

**What are the 5 primary taste sensation?**

**Primary Taste sensations**

1. Sweet
2. Salt
3. Sour
4. Bitter
5. Umami

**Write the name of enzyme that are used in glucometer?**

There are two types of enzymes that are used in Glucometer for the detection of glucose

1. Glucose Oxidase
2. Glucose Dehydrogenase

**Explain the two different working plate form of glucometer?****Wet Platform**

- I. The disposable Test strip is a wet platform.
- II. Blood travels through Microfluidic channel.
- III. Glucose oxidase is immobilized on the test strip.
- IV. Electrodes are fixed on strip and in contact with the enzyme.

**Dry Platform**

The Dry platform is consists of a

- ❖ A software
- ❖ A Hardware

It has a powerful processing unit with a

- Voltage converter
- Amplifier
- Filter
- Display unit

**On what basis Bio-FETs are classified?**

Bio-FETs are classified based on the bio recognition element used for detection:

- En-FET
- Immuno-FET
- DNA-FET
- CPFET
- Beetle/chip FET
- Artificial BioFET-based.

**What are the two main components of Bio-FET?**

A Bio-FET consists of two main components:

- Biological recognition element
- Field-effect transistor

**What is ceramic?**

Ceramics are inorganic non-metallic solid, they are hard, corrosion-resistant, brittle and cost effective. Example: Silicon Carbide.

**What is Troponin Complex?**

Troponin or the troponin complex is a complex of three regulatory proteins

- Troponin C
- Troponin I
- Troponin T

It is integral to muscle contraction in muscle. CTI and cTt are used as diagnostic and prognostic indicators in the management of AMI.

**What is Horse Reddish peroxidase function?**

HRP belongs to the ferroporphyrin group of peroxidases. The optimal pH is 6.0-6.5 with stability. Stable in the pH range of 5.0-9. HRP is a single chain polypeptide containing four disulphide bridges with Mol Wt. of about 44Kd with six Lysine residue suitable for conjugation with a protein.

**What is signal and biosignal?**

A signal is a function that conveys information about a phenomenon. The term biosignal is used for living organisms and their function that carries information about their behavior or state". Biosignals are the key to determine the functions in living organism.

**What is pressure sensing mechanism?**

Pressure sensing Elements and Basic Mechanism

- Pressure is sensed by mechanical elements such as plates, shells, tube and diaphragms.
- These elements are designed and constructed to deflect when pressure is applied.
- The basic mechanism in the pressure sensors is to convert the pressure to physical movement.

**What are Thermoreceptors?****Thermoreceptors (temperature sensors)**

A type of exteroceptors in which free nerve endings located in:

- Dermis of the skin
- Skeletal muscles
- Hypothalamus
- Liver

Cold receptors are high in number than warm receptors.

**What is Antibody affinity?**

Antibody affinity is the strength with which the epitope binds to an individual paratope (antigen-binding site) on the antibody. High affinity antibodies have following properties.

- They bind quickly to the antigen.
- They permit greater sensitivity in assays.
- They maintains this bond more readily under difficult conditions.

### Write parameter of biosensor?

#### Parameters

1. Sensitivity
2. Linearity
3. Selectivity
4. Range
5. Stability
6. Reproducibility
7. Response time

### What is Umami?

Umami is a pleasant taste characteristics of beef & chicken broth.

### What is BRET assay?

Bioluminescence resonance energy transfer (BRET), results from a naturally occurring phenomenon involving

- Resonance energy
- Enzymatic donor and
- Fluorescent protein acceptor.
- Range 1-10 nm range

BRET an ideal platform to study bio-molecular interaction or physiological processes in living cells.

### Advantages of Miniaturization?

Reduction in 10 times size would leads to

- 100 times reduction in heat flow.
- 100 times greater power loss.
- Linear size of electrodes 1000-time reduction in Potential energy
- In radius 10000 times reduction in volumetric flow.

### Properties of support material?

**Mean particle size:** It is the parameter which decides the porosity of a carrier.

**Hydrophilic character:** It determines the level of activity of an immobilized enzymes.

**Mechanical strength:** A mechanically strong material is required as a support material for immobilization.

### Electrochemical properties of DNA bases?

### Electrochemical properties of DNA bases

All four bases of DNA can be oxidized and produce their oxidation signals at highly positive potentials on carbon electrodes. The one electron reduction potentials of DNA nucleosides are

- Guanine 1.29
- Adenine 1.42
- Cytosine 1.6
- Thymine 1.7

### What is limit detection? Give example.

The minimum amount that a biosensor can detect is defined as its limit of detection (LOD). Example Glucometer.

### Principle of optical biosensor?

In optical biosensors, the interaction between the AB (BRE) and AG (Analyte) is marked by formation of AB-AG complex, leading to a change in the optical properties of the wave.

1. Reflectance
2. Absorbance
3. Refractive index
4. Optical path
5. Emission of Light.

### Hexokinase Based Fluorescent Biosensor?

- Fluorescent method for the determination of ADP.
- A fluorophore compound diethylamino-cumarin and genetics engineering techniques were used for the development of a biosensor for ADP.
- Biosensors shows high fluorescent binding with ADP.

### What are the two types of Photoreceptors?

Two types of Photoreceptors

#### Rods

- Very light sensitive
- Enable us to see in dim light

#### Cones

- Provide us with colour vision
- Their pattern of stimulation provide colour vision

### Light emitting diode and types?

- The light emitting Diodes (Transistors) Electrons (n-type) in the semiconductor recombine with electron holes (p-type), release energy in the form of photons.
- Colour of the emitted light is dependent on the material used for doping.
- In LEDs current can only flow in one direction.

- LEDs never burn out unless their current limit is passed.

### **Types of Light Emitting Diodes**

1. Gallium Arsenide infra-red
2. Aluminium Gallium phosphide-green
3. Gallium Indium Nitride (GalnN) - near ultraviolet, bluish-green and blue
4. Aluminium Gallium Nitride (AlGaN) - ultraviolet
5. To white LED light blue. LED is coated with phosphorous.

### **Immobilization of nucleic acids on the surface of biosensor?**

#### **DNA Biosensor**

DNA or RNA are the biorecognition elements in Nucleic acid Biosensors. The detection methods are based on recognition of complementary nucleic acid sequence

- A. Rapid
- B. Simple and
- C. Economical testing for clinical environmental, food analysis and other fields.

#### **Nucleic Acids immobilization**

The DNA fragments have to be immobilized in a way that retains their

- Stability
- Reactivity
- Accessibility

To target analyte for hybridization

#### **Adsorption**

DNA probes are immobilized by electrostatic adsorption between a negatively charged phosphate group of DNAs on the positively charged surface.

### **What is the function of the troponin complex?**

#### **Troponin Complex Function**

- **Troponin C** binds to calcium ions to produce a conformational change in Tnl.
- **Troponin T** binds to tropomyosin, interlocking them to form a troponin-tropomyosin complex.
- **Troponin I** binds to actin in thin myofilaments to hold the troponin-tropomyosin complex in place.

### **Define Signal and Bio signal?**

#### **Signals**

A signal is a function that conveys information about a phenomenon.

#### **Biosignals**

The term biosignal is used for living organisms and their function that carries information about their behavior or state. Biosignals are the key to

determine the functions in a living organism.

### **Write the application of the Zinc Oxide elements ?**

ZnO nano particles has application in many fields. Large surface area relative to their size :

- Medical Science
- Nano generator
- Gas sensor
- Solar cells
- Photodetector
- Biosensors

### **Applications of biosensors in environment?**

The increasing number of potentially harmful pollutants in the environment demands the use of

- Fast
- Cost-effective
- Analytical techniques
- Environmental protection
- Heavy metals/Phenols
- Nitrogen compounds

### **What are the advantages of the environmental biosensors?**

Environmental biosensors provide benefits over conventional analytical techniques.

- Portability
- Work on-site
- Measuring pollutants in complex matrices with minimal sample
- No pre-treatment

### **Thermal bio signals?**

Thermal bio signals are related to heat absorption or the release of heat. Both of these actions leads to increase or decrease in temperature of the system. The phenomena are the indication of a physiological change. A change in the temperature is a signal for the change.

### **Working of Electrochemical biosensors?**

Electrochemical biosensors have electrodes which translate the chemical signal into an electrical signal. Electrochemical sensors are able to detect various biomolecules in the human body such as glucose, cholesterol, uric acid, lactate, DNA, hemoglobin, blood ketones, and others.

### **Why too much demand of Biosensors in medical sciences?**

Because they Provides fast, accurate and Lab-quality results handheld device to accelerate decision making process in the medical field.

**What is hybridization?**

Hybridization is the process in which two complementary single-stranded DNA and/or RNA molecules bond together to form a double-stranded molecule.

**How biosensor detect water shortage in plants?**

Biosensor are very helpful in identifying the nutritional deficiencies and drought conditions in plants. A biosensor is designed to detect the drought conditions in plants by fabricating CNTs based electrode and stomata cells. The sensor used the opening and closing of stomata to detect the water condition in the plant. An ink was prepared by mixing CNTs with SDS and it was printed on the leaf surface. This ink was used an electrode, it create an electronic circuit.

**What are Proprioceptors?**

Proprioception also referred to as kinaesthesia, is the sense of self-movement, force, and body position. It is sometimes described as the "sixth sense". It is a sensory receptor which receives stimuli from within the body, especially one that responds to position and movement.

**Write down the features of glassy carbon?**

Glass-like carbon, (non- graphitizing), combines glassy and ceramic properties with those of graphite.

- Mechanically stable Impermeable to gases and fluids
- Chemically resistant
- High porosity
- Low background current
- Its low sensitivity to peroxide as well as to other mediators

**What is FET?**

The field-effect transistor also called FET is a type of transistor that uses an electric field to control the flow of current in a semiconductor. FETs are devices with three terminals: source, gate, and drain.

**What is Support matrix?**

Support matrix in a Biosensor is a sensitizing platform on which the Biorecognition process take place. It is required to be, stable, semi conductor and inert.

**Define Field effect transistor?**

The field-effect transistor is a type of transistor that uses an electric field to control the flow of current in a semiconductor. FETs are devices with three terminals: source, gate, and drain.

**How yeast biosensor works?**

The majority of yeast-based biosensors use a reporter gene under the control of an inducible promoter. The promoters are activated in the presence of a ligand of interest, either directly by the ligand/receptor dimer or via a signalling cascade.

**How bio signals are classified?**

Biosignals can also be divided in two main groups according to their source for measurement: Active and Passive.

**What are the characteristics of optical sensor techniques?**

Optical sensors have the ability to detect light, typically at a specific range of the electromagnetic spectra (ultraviolet, visible, and infrared). The sensor detects either wavelength, frequency, or polarization of light and converts it into electric signal due to photoelectric effect.

**What are Olfactory sensors?**

Olfactory receptors (ORs), also known as odorant receptors, are chemoreceptors expressed in the cell membranes of olfactory receptor neurons and are responsible for the detection of odorants for example, compounds that have an odour which give rise to the sense of smell.

**What do you know about CNTs?**

A carbon nanotube is a small cylindrical carbon structure made out of graphene. A carbon nanotube is a tube made of carbon with diameters typically measured in nanometres. Single-wall carbon nanotubes are one of the allotropes of carbon, intermediate between fullerene cages and flat graphene, with diameters in the range of a nanometre. The structure of CNT consists of enrolled cylindrical graphitic sheet (called graphene) rolled up into a seamless cylinder with diameter of the order of a nanometer. CNTs can act as antennas for radios and other electromagnetic devices. Conductive CNTs are used in brushes for commercial electric motors. They replace traditional carbon black. The nanotubes improve electrical and thermal conductivity.

**What are Quantum dots?**

Quantum dots (QDs) are man-made nanoscale crystals that that can transport electrons. When UV light hits these semiconducting nanoparticles, they can emit light of various colours. These artificial semiconductor nanoparticles that have found applications in composites, solar cells and fluorescent biological labels.

**Explain Cantilever biosensor?**

Cantilever biosensors are ultra-high sensitive electromechanical sensors that have been successfully used for label-free detection of a large number of biological entities. They are emerging as a technology that appears attractive for high-throughput drug discovery applications. Cantilever is functionalized by depositing a bioreceptor layer (top); surface-stress induced deflection upon binding between target analyte and bioreceptor (bottom).

**Describe Graphene?**

Graphene is an allotrope of carbon in the form of a single layer of atoms. It has in a two-dimension and has hexagonal lattice. In the structure one atom forms each vertex. It is the basic structural element of other allotropes, including graphite, charcoal, carbon

nanotubes and fullerenes. Graphene is the basic unit for the production of carbon based different nanostructures. Graphene, fullerene and carbon nanotubes are the allotropic forms of the carbon. Graphene is used sensors, faster and efficient electronics,, faster DNA sequencing, drug delivery and more.

### **Define Pressure sensor?**

A pressure sensor is a device for pressure measurement of gases or liquids and piezo electric materials (solids). Pressure sensors can measure other variables such as fluid/gas flow, speed, water level, and altitude. Also called as pressure indicators, piezometers and manometers, etc

### **POC testing?**

Point-of-care testing is defined as medical diagnostic testing at or near the point of care that is at the time and place of patient care.

### **Explain mass spectrometry?**

Mass spectrometry is an analytical tool useful for measuring the mass-to-charge ratio ( $m/z$ ) of one or more molecules present in a sample. These measurements can often be used to calculate the exact molecular weight of the sample components as well. The basic principle of mass spectrometry (MS) is to generate ions from either inorganic or organic compounds by any suitable method, to separate these ions by their mass-to-charge ratio ( $m/z$ ) and to detect them qualitatively and quantitatively by their respective  $m/z$  and abundance.

### **Application of nucleic acid biosensor?**

- Nucleic acid biosensors can be used to detect DNA/RNA fragments or either biological or chemical species. In the first application, DNA/RNA is the analyte and it is detected through the hybridization reaction this kind of biosensor is also called a Geno sensor.
- Clinical diagnostics of inherited diseases and pathogenic infections.
- Laboratory analysis like probe development for different hybridization techniques and microarray systems.

### **Write the Affinity Chromatography?**

Affinity chromatography is a separation method based on a specific binding interaction between an immobilized ligand and its binding partner. Examples include antibody/antigen, enzyme/substrate, and enzyme/inhibitor interactions. The principle of affinity chromatography is that the stationary phase consists of a support medium e.g. cellulose beads on which the substrate or sometimes a coenzyme has been bound covalently, in such a way that the reactive groups that are essential for enzyme binding are exposed.

### **What is the pressure sensing and its mechanism?**

#### **Pressure sensor**

A pressure sensor is a device for pressure measurement of gases or liquids.

**Mechanism of Pressure sensor**

The Pressure Sensor Mechanism is designed to measure or monitor tactile pressure. It is based on passive Radio Frequency Identification (RFID) sensor tags and is applicable to a variety of systems.

**Types of medium for laser production?**

- Liquid (dye lasers)
- Gas (gas laser)
- Solids (solid-state lasers and diode lasers)

**Define immuno-sensors?**

Immuno-biosensor is a type of biosensor to detect the formation of an immuno-complex using an antibody or antigen as a bioreceptor. Immunosensors are widely used and are particularly important for fast diagnosis of diseases in remote environments as well as point-of-care devices.

**Component of affinity based biosensor design?**

Affinity-based biosensors combine the high specificity of biorecognition agents such as ssDNA, antibodies and aptamers.

**Explain the Metallo-regulatory proteins?****Metallo-regulatory proteins**

The Metallo regulators are proteins that bind metals. A panel of metallo-regulatory proteins controls the expression of genes panel encoding

- Membrane transporters
- Metal trafficking proteins

These metal sensors manage

- Metal homeostasis
- Metal resistance.

**Functioning of metallo proteins**

The metal ions after binding with the gene promoter can induce different conformational changes

- Activates the promoter
- Inhibits the promoter

The binding could distort the promoter structure and can convert a poor promoter to a strong one.

**Electrical properties of Zinc oxide?**

Electrical Properties oxide has a relatively large direct band gap of around 3.3 eV at room temperature. The advantages associated with large band gap include :

- Higher breakdown voltage
- Ability to sustain large electrical field

- Lower electronic noise
- Lower temperature operation
- High power operation

### **Leakage population?**

There are at least two populations of the covalently attached enzymes

- Liable to Leakage
- Resistance to cleavage

The proportion of the two populations varies with the extent of cross linking during CNBr activation.

### **What is Escalating?**

Escalating means increasing rapidly. The escalating cost of healthcare.

### **Write growing areas of Biosensors?**

Biosensor technology is a fast growing technology for the detection of plant pathogens, in agriculture and food sector.

### **Size dependent properties?**

Every object has a set of size dependent physical properties. Size dependent physical properties change with the size. **Examples:** Mass, Volume.

### **How Lactate Oxidase Biosensor work?**

#### **Lactate Oxidase Biosensor**

- It utilizes lactate oxidase and has applicators in sports medicine.
- It is used to monitor lactate production during exercise. Results in just 10 seconds.
- Pocket-sized.
- It is simple, accurate, reliable lactate analyser for sports performance monitoring.

### **Advantages of Enzyme based Biosensor?**

#### **Advantages of Enzyme based Biosensor**

- Glucometer works by analyzing a small amount of blood.
- Glucometer is a small hand-held device.
- Glucometer works at ambient conditions.
- No pre-treatment of blood sample is required.
- Commercially available.

### **What is Lactate oxidase?**

The enzyme lactate oxidase catalyzes the oxidation of L-lactate to pyruvate in the presence of dissolved oxygen and it forms hydrogen peroxide. Lactate oxidase catalyzes the oxidation of lactic acid to pyruvate and hydrogen peroxide.

### **Source of Light?**

There are two sources of CO are used optical sensors.

- Natural sources include all the electromagnetic radiations such as visible, UV, IR and others.
- Man-made light sources include LASER and Light emitting Diodes.
- The light source is used according to the requirement.

### **What is DNA hybrid and PNA hybridization?**

#### **DNA hybridization**

DNA hybridisation is a process that is commonly used in medical diagnostic tests and genetic screening. In DNA hybridisation two complementary single-stranded DNA molecules combine through base pairing to form a single double-stranded DNA molecule.

#### **PNA Hybridization**

PNAs hybridize to complementary DNA or RNA in a sequence-dependent manner, according to the Watson–Crick model.